Nathan Arimilli, Shahmir Javed, Joseph Bailey

# Project 2 Memo: Minimum Variance Portfolio Optimization

## I. Introduction

The goal of this project is to evaluate the performance and stability of minimum variance portfolios constructed using the traditional sample covariance matrix and the Ledoit-Wolf shrinkage estimator for covariance. This analysis addresses a fundamental issue in portfolio optimization where sample covariance matrices estimated from historical data often produce unstable and noisy estimates, leading to poor out-of-sample performance. Using CRSP monthly stock returns accessed via WRDS, we implemented a robust, user-interactive Python program that performs covariance estimation, constrained quadratic optimization for portfolio weights, rolling window out-of-sample backtesting, turnover and stability diagnostics, and performance evaluation. The overall objective is to empirically assess whether the Ledoit-Wolf estimator, known for its stability and reduced estimation error, results in more robust and implementable portfolios compared to the traditional sample covariance matrix.

For this demonstration, we utilized our default parameters of retrieving data from 2010 to 2024. The actual backtest runs from January 2014 through December 2024, with the earlier data (2010-2013) used for the initial estimation windows. Our system automatically calculates when to start the backtest based on having sufficient historical data for the first estimation period and we allow users to specify any 6+ month time period from 2000 onwards. Our framework provides full flexibility in asset selection, but for our demonstration we used our default portfolio of 20 blue-chip stocks spanning six major sectors: technology (AAPL, MSFT, GOOG, IBM, NVDA), financials (JPM, BAC, MS, GS), healthcare (JNJ, PFE, MRK), consumer goods (KO, PG, WMT, HD), industrials (GE, CAT, MMM), and energy (XOM). Though users can input any combination of tickers available in the CRSP database, this default selection ensures our analysis captures various market dynamics and correlations.

## II. Methodology and Implementation

Our portfolio optimization framework employs a systematic approach for data acquisition, validation, and optimization. First, we established a data pipeline connecting to the WRDS CRSP database to retrieve monthly return data. As said before, we used our default setting of retrieving data from 2010 to 2024 in our example demo, with the extra years before the backtest period providing initial estimation data. Users can specify any start and end years, with our program requiring a minimum 3-year period for meaningful analysis. Second, we implemented a comprehensive data validation system where we applied the default requirement of 90% data availability (maximum of 10% missing data) within each estimation window. Users can adjust this threshold based on their data quality preferences. We use a 36-month rolling estimation window with monthly rebalancing, but users can specify any estimation window from 6 to 120 months. At each rebalancing date, we estimate covariance matrices using both methods. The sample covariance is calculated directly from the historical returns using the standard formula. The Ledoit-Wolf estimator, implemented via scikit-learn, optimally combines the sample covariance with a shrinkage target (scaled identity matrix) where the shrinkage intensity is determined analytically to minimize expected squared error. For each period, we solve the minimum variance optimization problem using constrained quadratic programming via cvxpy.

The optimization minimizes portfolio variance ($w^T \sum w$) subject to full investment constraints

(weights sum to 1) and position limits. Our program uses the default unconstrained settings allowing short positions up to -100% and long positions up to 100%, but this specification is flexible and up to the user. We employed multiple solver backends (OSQP, ECOS, SCS) sequentially to ensure robust convergence. The out-of-sample performance is then evaluated by applying the optimized weights to the following month's returns, creating a realistic assessment of each method's effectiveness.

## III. Parameter Choices and Rationale

As stated briefly before, our program offers complete parameter customization, but, for this demo, we utilized carefully chosen default values based on our intuition. The default 36-month estimation window balances statistical significance with market regime relevance. Shorter windows would increase responsiveness but suffer from greater estimation error, while longer windows risk incorporating outdated correlation structures. The default setting permitting selling up to -100% per asset with no explicitly long position constraints beyond the natural 100% limit when fully invested allows the optimizer to fully exploit negative correlations and construct efficient minimum variance portfolios, though it does increase complexity and turnover. The default risk-free rate of 4.2% annually reflects current 10-year Treasury Note yields as of late 2024, providing a realistic benchmark for Sharpe ratio calculations, though users should update this to reflect their current market environment. Our default data quality parameters require minimum 36 observations per stock (matching the estimation window) and maximum 10% missing data tolerance, though these can be adjusted based on data availability and quality requirements. The monthly rebalancing frequency represents a practical compromise between adaptation speed and transaction costs. The resulting 132 backtest periods from January 2014 to December 2024 encompass multiple market cycles including the post-2008 recovery, COVID-19 crisis, and recent inflationary period, ensuring robust evaluation across diverse market conditions.

## IV. Challenges Encountered

This project required addressing several technical and methodological challenges to ensure robust portfolio optimization across all market conditions. First, numerical instability in the sample covariance matrices posed significant problems, particularly during volatile periods. The sample covariance matrix occasionally approached singularity when correlations spiked, leading to optimization failures. Our code addressed this through sequential solver fallbacks (OSQP, ECOS, SCS), ensuring at least one solver could find the solution even with ill-conditioned matrices.

Secondly, data quality and synchronization required careful handling across the 20 stocks with potentially different trading calendars. We use a forward-fill methodology to handle minor data gaps while maintaining a strict validation routine requiring at least 90% data availability (our default setting) for the overall period and 75% within each estimation window.. Additionally, we queried the user for the test period before the tickers input so we can validate if enough data was available for that period (based on our default benchmark of 90%). If the program detects any stocks failing this threshold, we then requery the user to prevent stale data from distorting covariance estimates. The validation process confirmed all 20 default tickers had sufficient data coverage, though the system includes interactive replacement options for tickers with insufficient data.

Third, extreme portfolio weight instability with the sample covariance method created practical implementation challenges. The results show average monthly turnover exceeding 100%, with positions frequently flipping from large long to large short exposures (as seen in the final weights where JPM swings to +50.9% while PG goes to -45.7%). This instability stems from the fundamental problem of estimating 210 unique covariance parameters from just 36 monthly observations, making the optimization hypersensitive to small changes in recent returns. The Ledoit-Wolf method directly addresses this through shrinkage which reduces average turnover.

## V. Results Analysis

Our demo using the specified default parameters revealed substantial differences in risk-return profiles between the two methods. The sample covariance method achieved a total return of 315.79% over the 11-year backtest period compared to 242.30% for Ledoit-Wolf, translating to annualized returns of 13.83% versus 11.84% respectively. This approximately 2% annual return advantage for the sample covariance method came at the cost of elevated risk with an annualized volatility of 18.67% for this method versus 14.03% for the Ledoit-Wolf method, representing a 33% increase in volatility. The 12 month rolling volatility is visualized in Figure 1 in the appendix and clearly illustrates this difference, with the sample covariance volatility frequently spiking above 30% during stress periods, particularly from 2022-2024, while the Ledoit-Wolf volatility remained consistently below 25%. This volatility differential is further confirmed by the maximum drawdown of -30.88% for sample covariance versus -19.20% for Ledoit-Wolf, demonstrating the latter's superior downside protection. Additionally, despite lower absolute returns, Ledoit-Wolf achieved a slightly superior Sharpe ratio of 0.576 versus 0.570 for sample covariance, indicating marginally better risk-adjusted performance. More notably, the Sortino ratio of 0.930 for Ledoit-Wolf versus 0.777 demonstrates significantly better downside risk management, which is a crucial consideration for risk-averse investors.

Moreover, the turnover analysis reveals perhaps the most striking difference between the methods. The sample covariance portfolio exhibited median monthly turnover of 87.6%, nearly four times higher than Ledoit-Wolf's 23.4%. The average figures are even more dramatic with 106.7% for sample covariance versus 25.2% for Ledoit-Wolf. The monthly portfolio turnover visualization in Figure 2 shows sample covariance turnover regularly exceeding 100% with spikes reaching 375.9%, while Ledoit-Wolf's maximum reached 83.0%. These turnover differences have profound implications for net returns after transaction costs. Charging a cost per unit of turnover, the annual drag is roughly twelve times that cost times the average monthly turnover. At a modest 10 basis points per unit of turnover, the sample covariance portfolio would give up about 1.3% per year versus roughly 0.3% per year for Ledoit-Wolf, a differential near 1% that erases close to half of the sample method's 2% gross return advantage. That advantage disappears entirely around 20 basis points, beyond which Ledoit-Wolf is superior on a net basis as well. Even before that point, Ledoit-Wolf already dominates on volatility, drawdown, downside risk, and turnover, so the after-cost case favors the shrinkage portfolio.

Additionally, the portfolio weight evolution visual in Figure 3 reveals extreme instability in the sample covariance methods. This visual shows major oscillations over the backtest period, with JPM, for example, reaching -50% of portfolio weight in around 2016 to nearly 100% in 2022. In the final period, the sample covariance has JPM at 50.9% long and PG at -45.7% short, demonstrating extreme concentration risk. In contrast, Ledoit-Wolf's largest positions were GOOG at 20% long and CAT at -18% short, representing more balanced and implementable allocations. The cross-sectional standard deviation of weights, seen in Figure 3, also quantifies this instability in the sample covariance method. The sample covariance method standard deviation of weights is significantly higher than the Ledoit-Wolf method, with this spread widening even more significantly post-pandemic. Furthermore, The Ledoit-Wolf shrinkage parameter evolution, visualized in Figure 1, provides insight into the method's adaptability. Shrinkage intensity varied between 0.15 and 0.30, peaking during the COVID-19 crisis as the algorithm recognized heightened estimation uncertainty. The gradual decline from 2022-2024 suggests improving estimation quality as markets stabilized.

Finally, the efficient frontier visualization for the final estimation period in Figure 4, reveals an interesting paradox. The sample covariance frontier actually shows lower volatility needed for a given return level compared to Ledoit-Wolf in the in-sample period. However, this apparent superiority doesn't translate to out-of-sample performance, highlighting the overfitting problem inherent in sample estimation. The Ledoit-Wolf frontier exhibits a more stable slope, suggesting more reliable risk-return trade-offs that better persist out-of-sample.

## VI. Conclusion

Overall, this analysis demonstrates that while sample covariance optimization can generate higher absolute returns, the Ledoit-Wolf shrinkage method led to a significant reduction in turnover, volatility, and maximum drawdown, while increasing stability in weights and risk-adjusted return. These improvements stem from the fundamental difference in how the methods estimate covariance: while sample covariance uses only historical data (leading to overfitting and instability with limited observations), Ledoit-Wolf optimally combines the sample estimate with a stable shrinkage target, effectively reducing the multi-parameter estimation problem to a more manageable dimension. This shrinkage acts as a form of regularization, dampening the extreme correlations and spurious patterns that cause the sample covariance method to produce unstable, concentrated portfolios that change dramatically each period, particularly out-of-sample.

We utilized several resources to assist us in writing the code for this assignment including lecture material, slides, notes, and hand outs from FIN 684: Advanced Corporate Finance/Investment Theory as well as LLMs (e.g., ChatGPT, Claude).

We attest that this assignment has been completed in accordance with the academic integrity protocols and honor codes of the McCombs School of Business and the University of Texas. We relied on no unauthorized resources (including material submitted or produced by other students—past or present, individuals not listed below, internet message boards, solution manuals, or other forms of outside assistance). Any assistance from AI tools (e.g., ChatGPT) has been appropriately cited. This work product is the sole result of our efforts and is being submitted as such. (Nathan Arimilli na26249, Shahmir Javed sj29745, Joseph Bailey jb87575).

Figure 1:

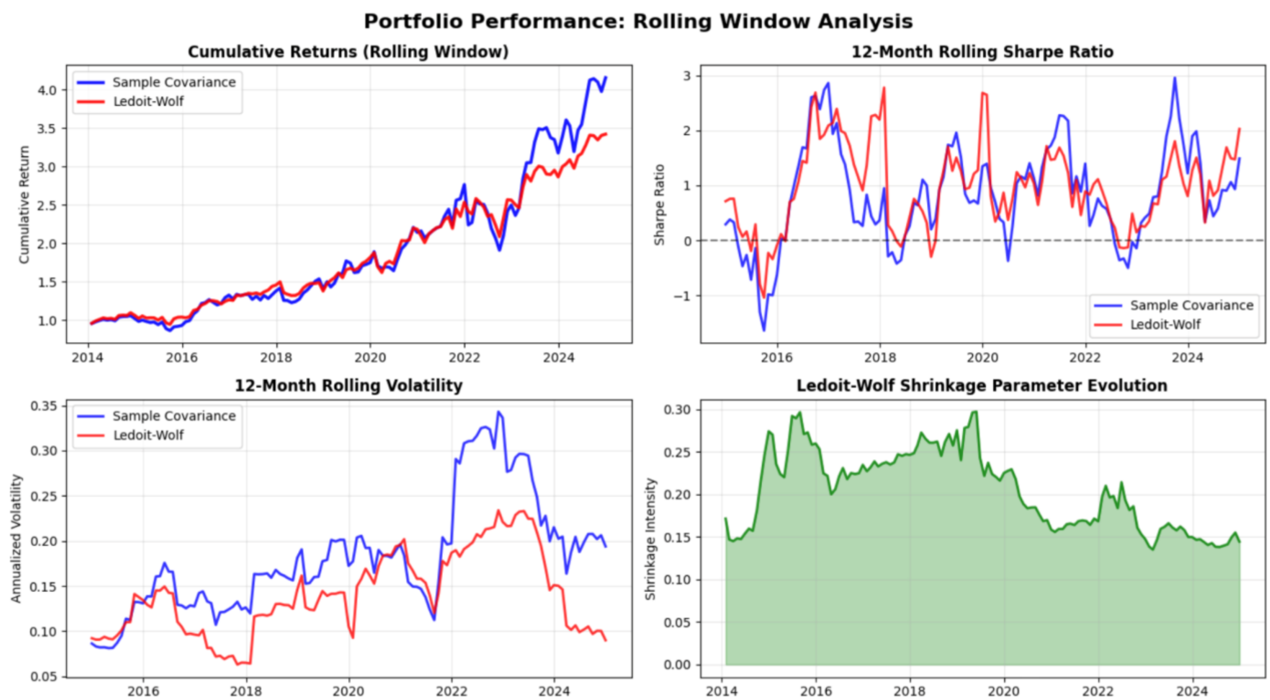



Figure 2:

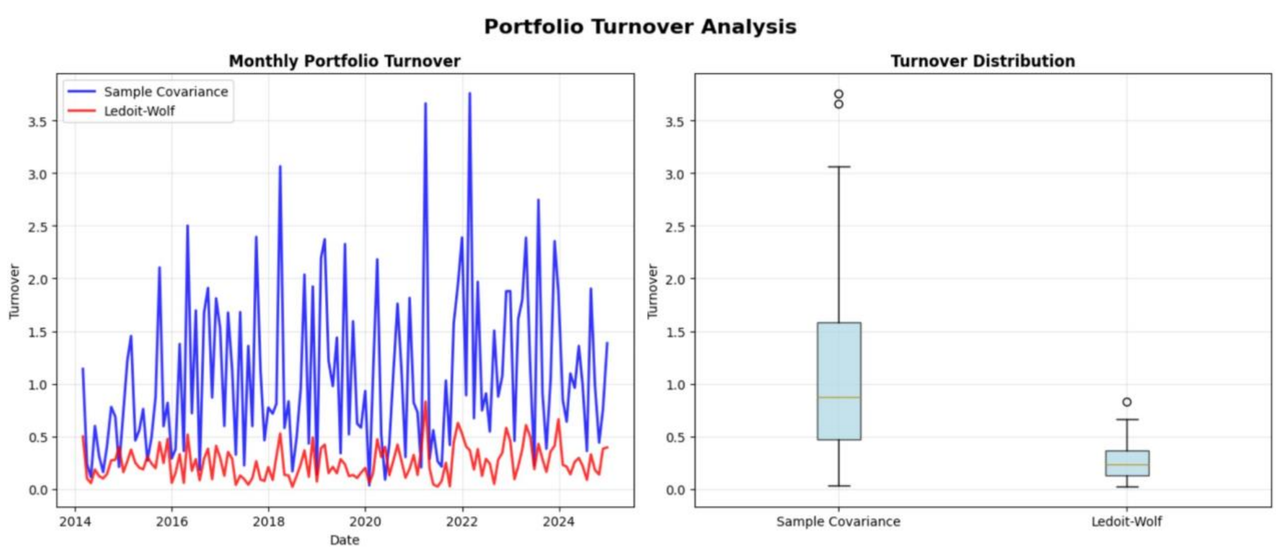



Figure 3:

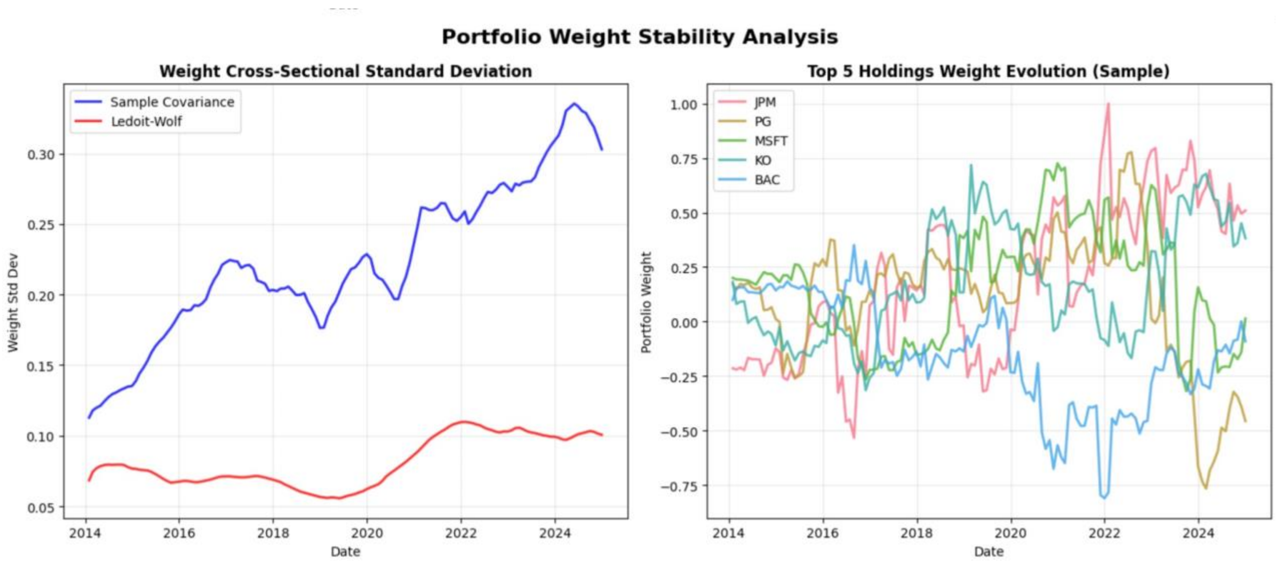
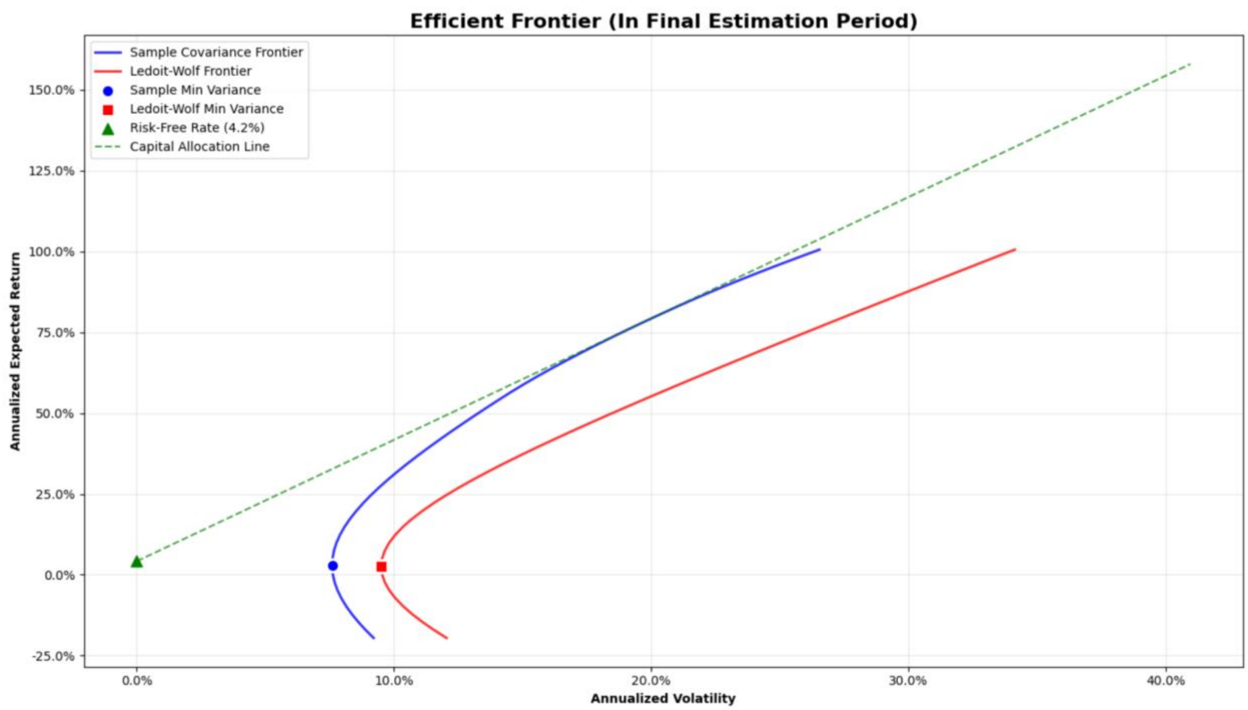

Portfolio Weight Stability Analysis
Weight Cross-Sectional Standard Deviation
Sample Covariance
Ledoit-Wolf
Weight Std Dev
Date
Top 5 Holdings Weight Evolution (Sample)
JPM
PG
MSFT
KO
BAC
Portfolio Weight
Date

Figure 4:


Efficient Frontier (In Final Estimation Period)
Sample Covariance Frontier
Ledoit-Wolf Frontier
Sample Min Variance
Ledoit-Wolf Min Variance
Risk-Free Rate (4.2%)
Capital Allocation Line
Annualized Expected Return
Annualized Volatility